MD. Mahmudul Islam

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Career Objective

AI/ML developer with hands-on experience building end-to-end systems seeking an entry-level role to apply skills in deep learning and applied NLP to solve real-world problems, while continuing to grow through production-focused engineering work.

Education

B.Sc. in Computer Science and Engineering (CSE)
Varendra University, Rajshahi, Bangladesh | Aug 2025
CGPA: 3.43

Projects

Skin Disease Detection & AI Advisor System | "Live Demo"

- Built an AI-powered skin disease detection system that classifies 10 diseases from skin images using **EfficientNet-B0** transfer learning achieving **84.24% validation accuracy**
- Integrated **Groq LLaMA3-70b** LLM to generate real-time medical recommendations, next steps and tips based on detection results
- Developed a **REST API** using **FastAPI** with endpoints for quick and full analysis, with auto-generated API documentation
- Handled **class imbalance** using **weighted loss function** improving model fairness across all 10 disease classes
- Built an interactive demo UI using **Gradio** and containerized the application using **Docker** for easy deployment

Adaptive Document Preparation System | "Live Demo"

- Built a **CLI and web-based adaptive study system** that ingests a 50-page PDF into 10 sections using **PyMuPDF** and serves questions via a **Gradio** interface
- Integrated **Groq (LLaMA 3.3-70b)** to generate **MCQs** with adaptive prompting — injecting **SQLite Knowledge Base** history of weak topics into the **LLM prompt** each session
- Designed **weak-topic tracking logic** that identifies consistently wrong answers across sessions and focuses new question generation on those areas while excluding mastered content
- Demonstrated **3-iteration adaptive Scenario B** with progressive learning behavior proven end-to-end

Bengali Empathetic Conversational AI (LLaMA Fine-Tuning)

- Fine-tuned **LLaMA 3.1-8B-Instruct** using **LoRA** with **Unsloth** for memory-efficient training
- Implemented **4-bit quantization** and **gradient checkpointing** to train large LLM on **Kaggle GPU (T4)**
- Designed instruction-style prompt pipeline for Bengali conversational data (**8192 token context**)
- Achieved **perplexity: 2.56**, **validation loss: 0.94**
- Evaluated model using **BLEU** and **ROUGE** metrics; validated **empathetic response quality**
- Built **modular OOP pipeline** (**DatasetProcessor**, **FineTuner**, **Evaluator**)
- Ensured **experiment reproducibility** and end-to-end workflow in **Kaggle**

Undergraduate Thesis: Smartphone Addiction Rate Prediction of People and Their Impaired Mental Health

- Collected and preprocessed survey data from students to build a smartphone addiction prediction dataset.
- Analyzed correlation between smartphone usage patterns and mental health indicators using statistical methods
- Published results in Undergraduate Conference on **Intelligent Computing and System** (**UCICS 2025**), Varendra University, Rajshahi.

Technical Skills

- **Languages:** Python, SQL, C/C++
- **Frameworks:** FastAPI, Gradio, Typer, PyTorch
- **LLM & AI:** Prompt Engineering, LLM Integration(Groq, LLaMA) , Adaptive AI Systems, PDF Processing (PyMuPDF)
- **Machine Learning:** scikit-learn, Regression, Classification, Naive Bayes, Feature Engineering, EDA
- **Data Science Tools:** NumPy, Pandas, Matplotlib
- **Tools & Platforms:** Google Colab, Jupyter Notebook, Git, VS Code, Docker
- **Database:** MySQL, SQLite

Certifications:

- Machine Learning Specialization — DeepLearning.AI (Coursera)
- Mathematics for Machine Learning and Data Science Specialization — DeepLearning.AI (Coursera)
- LangChain Chat with Your Data(DeepLearning.AI)
- Introduction to Python — DataCamp

References

Available upon request.